

Separation of Variables

1 Heat Diffusion

- problem and model

2 Decoupling Space from Time

- reduction to ordinary differential equations
- the method of separation of variables
- imposing the boundary conditions

3 Fourier Series

- imposing the initial conditions
- plots to examine the convergence

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Industrial Math & Computation
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cooling an insulated tube at the ends

Imagine a tube filled with hot water.

The tube is insulated everywhere, except at the ends.

*At the uninsulated ends of the tube cooling happens,
what is the evolution of the temperature distribution?*

a mathematical model for heat diffusion

Let $u(x, t)$ be the temperature at position $x \in [0, 1]$ at time $t \geq 0$.

The heat diffusion is modeled by a partial differential equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2},$$

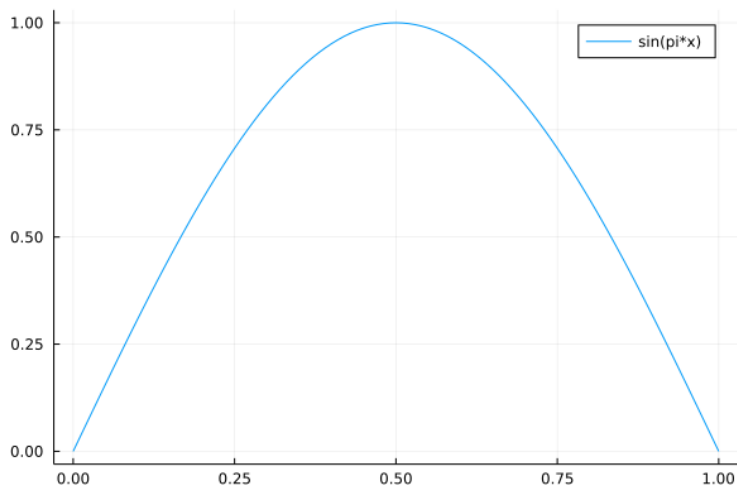
where

- $u(x, 0) = 1$ is the initial condition, temperature is everywhere 1,
- $u(0, t) = 0$ is the left boundary condition, temperature at zero,
- $u(1, t) = 0$ is the right boundary condition, temperature at zero.

Geometric interpretation:

The rate at which the temperature decreases is proportional to the concavity of the temperature distribution.

a sine profile of the temperature distribution



warming up

Throughout this lecture we consider a cooling off process.

Exercise 1:

Consider the heating of a tube filled with cold water.

The tube is insulated everywhere, with heat provided at the ends.

Answer the following.

- 1 Formulate the boundary conditions for this model.
- 2 What is the geometric interpretation?

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reduction to ordinary differential equations

Given is a partial differential equation in $u = u(x, t)$,
in two variables:

- 1 x specifies the position, *where* u is,
- 2 t is a moment in time, *when* u exists.

In an attempt to find a solution, we simplify the problem,
assuming position and time to be independent from each other.

This assumption implies that we can write u as a product of

- 1 a function in x , and
- 2 a function in t .

This method reduces the given partial differential equation in two
decoupled ordinary differential equations.

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the method of separation of variables

Assuming that x and t are independent from each other:

$$U(x, t) = T(t)X(x),$$

implies that we propose U , a product of two functions T and X , as the solution u to the partial differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$.

Now we impose the partial differential equation on $U(x, t)$:

$$\frac{\partial U}{\partial t} = \frac{\partial T}{\partial t}X(x) \quad \text{and} \quad \frac{\partial^2 U}{\partial x^2} = T(t)\frac{\partial^2 X}{\partial x^2}.$$

Then the original partial differential equation becomes

$$\frac{\partial T}{\partial t}X(x) = T(t)\frac{\partial^2 X}{\partial x^2}.$$

decoupling into ordinary differential equations

The imposition of $U(x, t) = T(t)X(x)$ on the equation led to

$$\frac{\partial T}{\partial t} X(x) = T(t) \frac{\partial^2 X}{\partial x^2}$$

and is rewritten as

$$\frac{\partial T}{\partial t} \frac{1}{T(t)} = \frac{1}{X(x)} \frac{\partial^2 X}{\partial x^2}.$$

The assumption that space and time are independent from each other implies the decoupling of the equation, wedging in a constant λ :

$$\frac{\partial T}{\partial t} \frac{1}{T(t)} = \lambda = \frac{1}{X(x)} \frac{\partial^2 X}{\partial x^2}.$$

The constant λ is independent of x and t .

two ordinary differential equations

Proposing $U(x, t) = T(t)X(x)$ as solution allows to wedge in λ :

$$\frac{\partial T}{\partial t} \frac{1}{T(t)} = \lambda = \frac{1}{X(x)} \frac{\partial^2 X}{\partial x^2}$$

which then leads to

$$\begin{aligned} \frac{\partial T}{\partial t} &= \lambda T(t) \quad \Rightarrow \quad T(t) = T(0)e^{\lambda t} \\ \frac{\partial^2 X}{\partial x^2} &= \lambda X(x). \end{aligned}$$

Because we model a cooling off process, we want decay:

$$\lambda = -\omega^2 < 0.$$

the eigenfunctions of the Laplacian operator

The choice of symbol for the separation constant λ suggest a connection with the eigenvalue problem.

Using the Laplacian ∇^2 , we abbreviate $\frac{\partial^2 X}{\partial x^2} = \lambda X(x)$ as $\nabla^2 X = \lambda X$.

The function $X(x)$ is an *eigenfunction* of ∇^2 corresponding to λ .

Exercise 2:

The Laplacian with zero boundary conditions is negative definite:

$$\text{For any } \phi \neq 0 : \langle \nabla^2 \phi, \phi \rangle < 0.$$

Prove this statement.

Hint: See the outline of exercise 11.6 in the text book.

solving for space

As a solution to

$$\frac{\partial^2 X}{\partial x^2} = -\omega^2 X(x),$$

consider a linear combination of a cosine and a sine function:

$$\begin{aligned} X(x) &= a \cos(\omega x) + b \sin(\omega x) \\ X'(x) &= -a\omega \sin(\omega x) + b\omega \cos(\omega x) \\ X''(x) &= -a\omega^2 \cos(\omega x) - b\omega^2 \sin(\omega x) \\ &= -\omega^2 X(x). \end{aligned}$$

Therefore, for some constants a and b ,

$$X(x) = a \cos(\omega x) + b \sin(\omega x)$$

is a solution to the second order differential equation.

the general form of a solution

Combining

$$T(t) = T(0)e^{-\omega^2 t}$$

and

$$X(x) = a \cos(\omega x) + b \sin(\omega x)$$

leads to

$$U(x, t) = T(t)X(t) = T(0)e^{-\omega^2 t} \left(a \cos(\omega x) + b \sin(\omega x) \right).$$

This solution depends on $T(0)$, ω , a , and b .

Let $T(0)a = A$ and $T(0)b = B$, then the general form becomes

$$U(x, t) = e^{-\omega^2 t} \left(A \cos(\omega x) + B \sin(\omega x) \right).$$

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imposing the boundary conditions

On the general form of a solution

$$U(x, t) = e^{-\omega^2 t} \left(A \cos(\omega x) + B \sin(\omega x) \right)$$

we impose the boundary conditions

$$u(0, t) = 0 \quad \text{and} \quad u(1, t) = 0.$$

The condition $u(0, t) = 0$ leads to

$$U(0, t) = \underbrace{e^{-\omega^2 t}}_{\neq 0} (A \cdot 1 + B \cdot 0) \Rightarrow A = 0.$$

So we simplified the form into $U(x, t) = e^{-\omega^2 t} \left(B \sin(\omega x) \right).$

finding values for ω

Imposing $u(1, t) = 0$ on $U(x, t) = e^{-\omega^2 t} (B \sin(\omega x))$ leads to

$$U(1, t) = \underbrace{e^{-\omega^2 t}}_{\neq 0} (B \sin(\omega)) \Rightarrow B = 0 \quad \text{or} \quad \sin(\omega) = 0.$$

Setting B to zero sets U to zero, contradicting $u(x, 0) = 1$, thus $B \neq 0$.

$$\sin(\omega) = 0 \Rightarrow \omega = n\pi, \quad n = 1, 2, \dots$$

As $B \neq 0$, for each multiple n of π , we set $B = B_n$. So, we have

$$U_n(x, t) = e^{-n^2 \pi^2 t} (B_n \sin(n\pi x)), \quad n = 1, 2, \dots,$$

and thus infinitely many solutions to the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$.

adding it all up

The operators $\frac{\partial}{\partial t}$ and $\frac{\partial^2}{\partial x^2}$ are linear. Therefore, consider

$$U_n(x, t) = e^{-n^2\pi^2 t} \left(B_n \sin(n\pi x) \right), \quad n = 1, 2, \dots,$$

as a basis of functions.

Our new general form of a solution is then

$$U(x, t) = \sum_{n=1}^{\infty} B_n \left(e^{-n^2\pi^2 t} \sin(n\pi x) \right).$$

The values B_n are the coefficients of the solutions in the basis of sine functions with decaying amplitudes.

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imposing the initial conditions

Imposing the initial condition $u(x, 0) = 1$ on

$$U(x, t) = \sum_{n=1}^{\infty} B_n \left(e^{-n^2 \pi^2 t} \sin(n\pi x) \right)$$

leads to

$$U(x, 0) = \sum_{n=1}^{\infty} B_n \sin(n\pi x) = 1.$$

To find the values for B_n ,
we exploit the orthogonality of the basis $\sin(n\pi x)$, $n = 1, 2, \dots$

Fourier series

We compute B_n in a Fourier series approximation to

$$1 = \sum_{n=1}^{\infty} B_n \sin(n\pi x).$$

The inner product

$$\left\langle \sin(n\pi x), \sin(m\pi x) \right\rangle = \int_0^1 \sin(n\pi x) \sin(m\pi x) dx = \begin{cases} 0 & \text{if } n \neq m \\ \frac{1}{2} & \text{if } n = m \end{cases}$$

shows that sine functions running at different frequencies are orthogonal to each other.

exploiting orthogonality of the basis

Applying the inner product

$$\left\langle \sin(n\pi x), \sin(m\pi x) \right\rangle = \int_0^1 \sin(n\pi x) \sin(m\pi x) dx = \begin{cases} 0 & \text{if } n \neq m \\ \frac{1}{2} & \text{if } n = m \end{cases}$$

to

$$1 = \sum_{n=1}^{\infty} B_n \sin(n\pi x)$$

leads to

$$\begin{aligned} \left\langle 1, \sin(m\pi x) \right\rangle &= \sum_{n=1}^{\infty} B_n \left\langle \sin(n\pi x), \sin(m\pi x) \right\rangle \\ &= B_m \frac{1}{2}. \end{aligned}$$

evaluating an integral

$$\begin{aligned}\frac{1}{2}B_m &= \left\langle 1, \sin(m\pi x) \right\rangle \\ &= \int_0^1 \sin(m\pi x) dx \\ &= \frac{1}{m\pi} \left(1 - \cos(m\pi) \right)\end{aligned}$$

The cosine is $+1$ at even multiples of π , and -1 at odd multiples:

$$\cos(m\pi) = \begin{cases} -1 & \text{if } m = 2k + 1, \\ +1 & \text{if } m = 2k. \end{cases}$$

For even m , $B_m = 0$. For odd m , $B_m = \frac{4}{m\pi}$ or $B_{2k+1} = \frac{4}{(2k+1)\pi}$.

the general solution

The imposition of $u(x, 0) = 1$ on

$$U(x, t) = \sum_{n=1}^{\infty} B_n \left(e^{-n^2 \pi^2 t} \sin(n\pi x) \right)$$

has given us

$$B_{2k+1} = \frac{4}{(2k+1)\pi}, \quad k = 0, 1, 2, \dots$$

which leads to the general solution

$$U(x, t) = \sum_{k=0}^{\infty} \frac{4}{(2k+1)\pi} \left(e^{-(2k+1)^2 \pi^2 t} \sin((2k+1)\pi x) \right).$$

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plots to examine the convergence

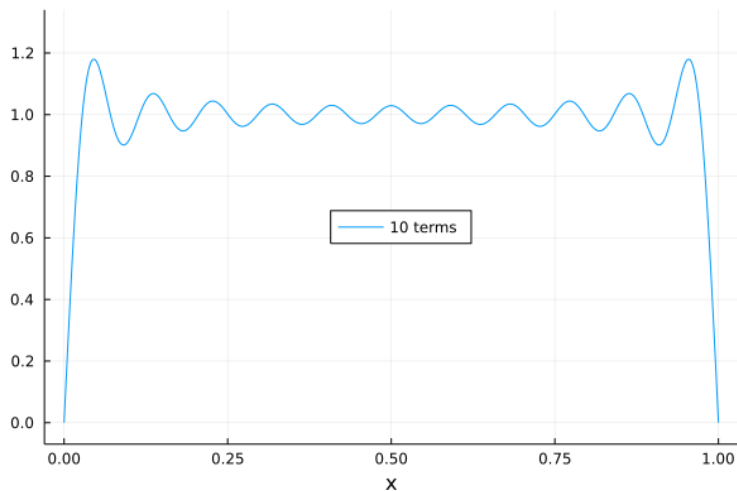
Looking at the general solution

$$U(x, t) = \sum_{k=0}^{\infty} \frac{4}{(2k+1)\pi} \left(e^{-(2k+1)^2 \pi^2 t} \sin((2k+1)\pi x) \right)$$

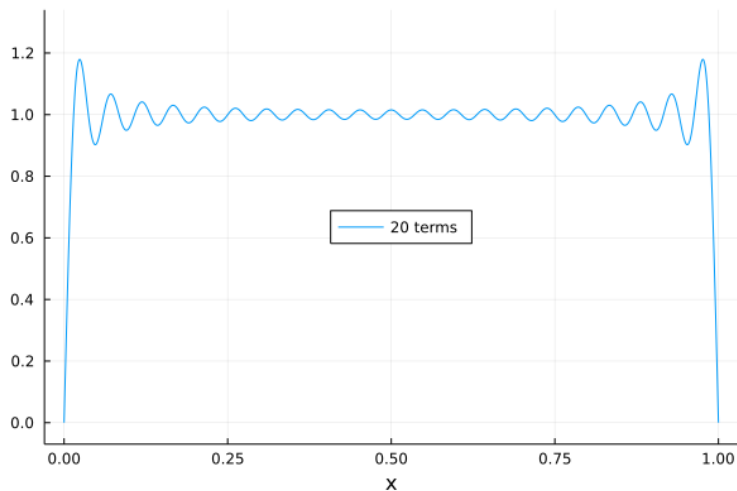
we ask two convergence questions:

- 1 How many terms are needed in the sum?
- 2 How fast does the temperature go to zero?

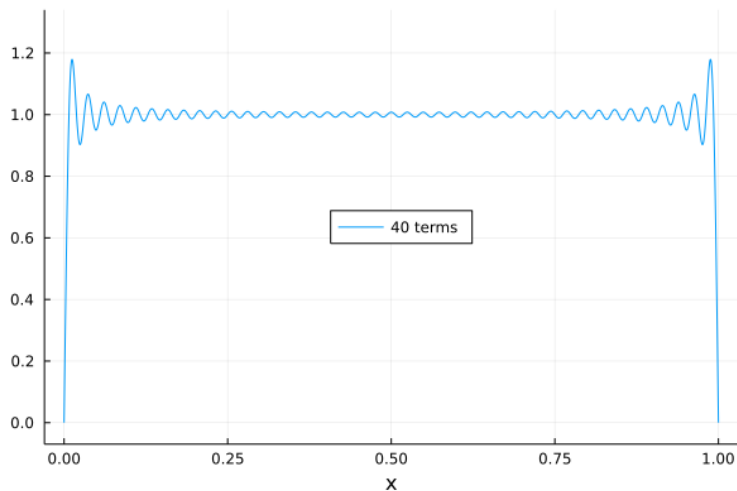
at $t = 0$ with 10 terms in the sum



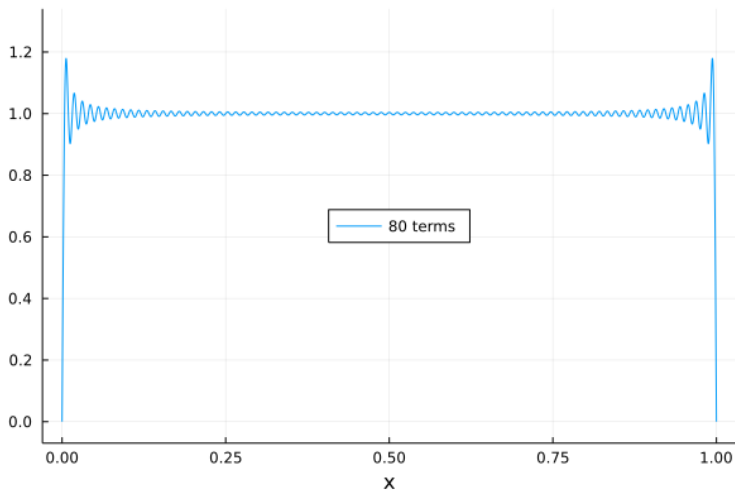
at $t = 0$ with 20 terms in the sum



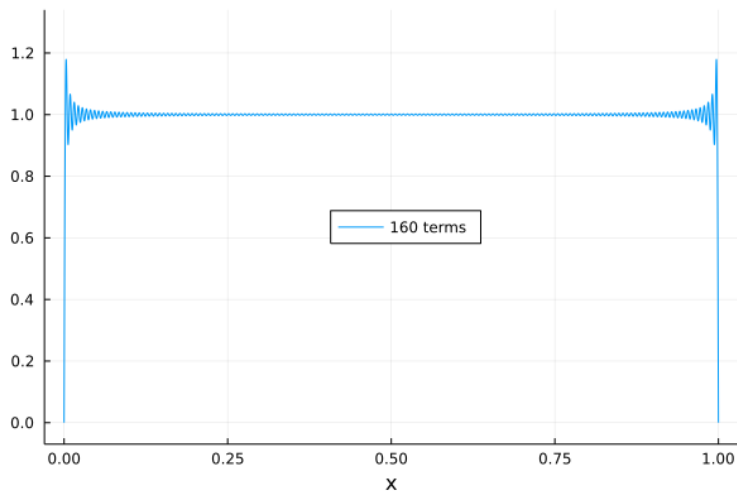
at $t = 0$ with 40 terms in the sum



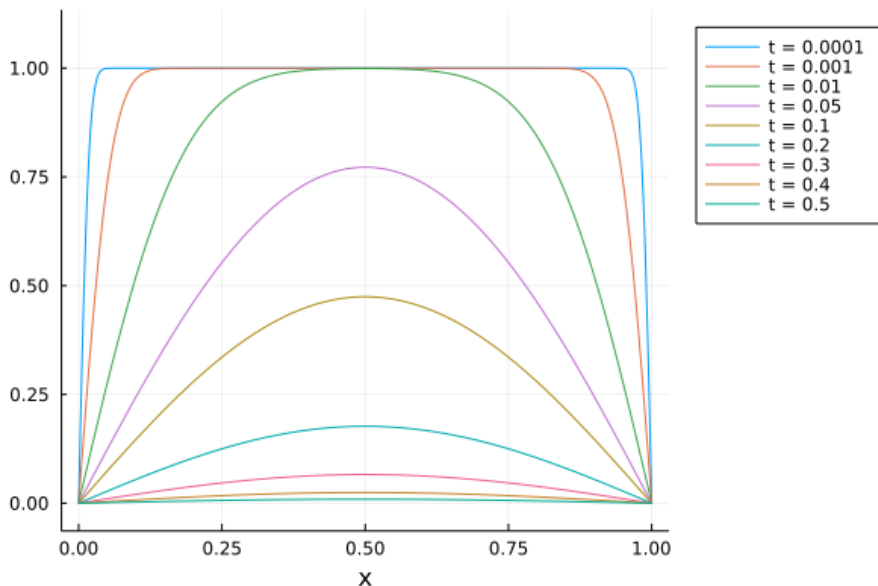
at $t = 0$ with 80 terms in the sum



at $t = 0$ with 160 terms in the sum



the evolution in time with 160 terms in the sum



a bell shaped heat profile

Exercise 3:

Instead of $u(x, 0) = 1$, consider a bell shaped profile, with

- 1 the center is at the middle $1/2$ of the interval $[0, 1] \ni x$, and
- 2 the standard deviation of the Gaussian is $1/2$ as well.

Do the following.

- 1 Set up the model for the evolution of the temperature distribution.
- 2 Compute the Fourier series of the profile.
- 3 Provide plots to display the evolution.

the Fourier ring problem

A long rod is bent and its ends welded to form a ring. Given an initial temperature profile around the ring, how will this temperature profile evolve over time?

Proposal for a Topic of a Project:

Study the solution of Problem 3 in our text book.

Do the following:

- Describe the solution using your own words.
- Do the Fourier series computations.
- Write a report on your computational experiments.

the heat diffusion problem in polar coordinates

Consider the preparation for a camping trip.
To not freeze to death during the night in a tent,
a heater for tent camping is needed.

Proposal for a Topic of a Project:

Our tent has a circular base, with the heater at its center.
Develop a mathematical model to prepare for a safe camping trip.

Do the following:

- Gather data on tent heaters.
- Describe the heat diffusion equation in polar coordinates.
- Interpret the computational results with the data.

summary and bibliography

The method of separation of variables transforms the solution of one partial differential equation into the solutions of infinitely many ordinary differential equations.

Fourier series were developed to study heat diffusion.

We covered Problem 1 of section 11.3 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

Separation of variables is covered in chapter 2 of the book

- Richard Haberman:
Applied Partial Differential Equations with Fourier Series and Boundary Value Problems. Pearson, 2013.